Unit 9 Passage 1
What is a Database?

A place to keep information

What is a database? Here is a simple answer: a database is a physical place (a "base") to keep information ("data"). However, databases today are much more complicated than that. Modern **digital** databases do not just store data. Databases also organize and manage the data, protect it, and make it quick and easy to access for users. To understand what a database is and how it works, we need to understand what **data** is.

So what is data?

The word "data" is the plural form of the word, "**datum**." We can think of a datum as a piece of information. A movie, a picture, a song, a book, a paragraph, a sentence, a word, and even a single letter or number, like '1' or '0', can all be considered to be information. We can divide that information in different ways. A book, a page of a book, a paragraph, or a letter can all be considered data at different levels of measurement. When we talk about data in a database, we reduce the data to the smallest measurable units, what are bits (1s and 0s).

Humans have created and managed information for thousands of years, and organizing and managing information has been a fundamental problem for all of human history. For example, imagine an old library with 200,000 books. If you wanted to find one specific book, how would you find it? Before computers, libraries kept information about each book, such as the title, author, and genre, on index cards. You would have to look through the cabinets of index cards to find on which shelf the book is located. This kind of system is an **analog** database.

Today, the use of computers has increased the amount of information available to us. While access to more information is good, it also makes the problem of organizing that information more difficult. To manage all that data, we created IT, or "Information Technologies." Digital databases are a key component of IT management. Databases do not only store information, like a folder or simple spreadsheet, but they also make the information easy and fast to access by using a database management system.

Database Management Systems

A database management system, often called a **DBMS**, is a program that organizes and processes the data. A basic DBMS uses spreadsheets to manage all the information stored in the database. The spreadsheet follows a **schema** by processing information into **tables**, which organize the columns as '**fields**' and the rows as '**records**'. A user can **query** the database to quickly access information. Many databases use a Structured Query Language, or **SQL**, to search for the user's query in the fields and records. In the early 2000s, computer scientists created NoSQL (meaning Not Only SQL) databases, which process information in a variety of ways. We will read more about different kinds of SQL and NoSQL databases in passage 2.

So, in conclusion, what is a database? A database is a physical storage for digital information that uses a DBMS to manage and access the information. Databases allow people and other computer systems to access information quickly and easily.

Vocabulary

Do your best to learn the following IT vocabulary. If you do not understand a word, look it up in the dictionary.

digital / **analog**
datum (singular) and **data** (plural)
Database Management System (**DBMS**)
Schema
table, **field** (column), **record** (row)
query
Structured Query Language (**SQL**)
Not Only Structured Query Language (**NoSQL**)